

Lecture 15

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Random Number Generators

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Introduction to Random Numbers

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- Many uses of random numbers in cryptography
 - ▣ Session keys
 - ▣ Public key generation
 - ▣ one-time pad encryption system
- Random Numbers are used in:
 - ▣ simulation whether discrete event, continuous, Monte Carlo, etc.
 - ▣ Cryptographic protocols – keys, challenges, etc.

Early Random Number Generators

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- *Early Approaches:* Methods were carried out by hand and used to create tables of numbers.
- Examples are dice, coin flipping.

Computer Generated Random Numbers

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- With the rise of computers, the focus shifted to numeric or arithmetic ways to generate random numbers.
 - ▣ Not truly “random,” since they are not unpredictable. The whole sequence can be determined using the formula and the input values.
 - ▣ Also called: pseudo random or “close-to” random
 - ▣ The goal is to produce numbers that appear to be random.

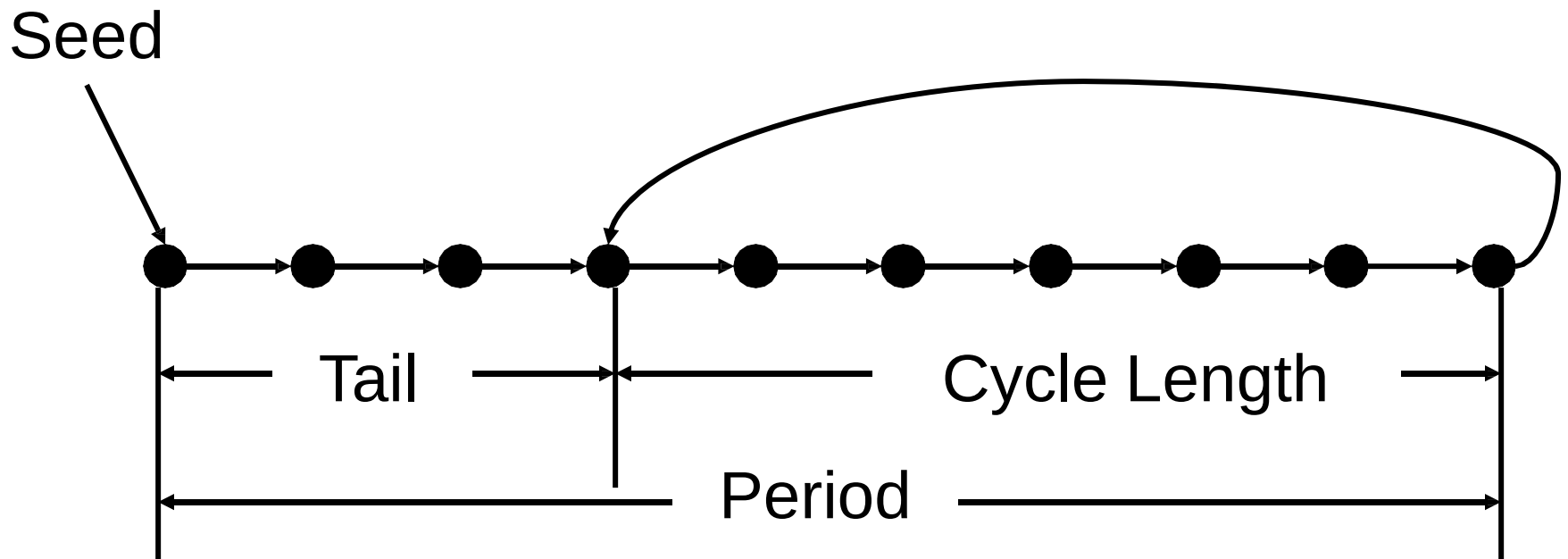
Computer Generated Random Numbers Basics

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- We will be discussing methodologies that will ultimately yield $U(0,1)$ distributions.
- All generators require a **seed** (X_0) to begin the random number sequence.
- The values are considered to be pseudo-random, since the generation of the values is deterministic and allows us to repeat results.
- $x_n = f(x_{n-1}, x_{n-2}, \dots)$
- We will be concerned with the statistical properties of **randomness** , **period** and **cycle length**.

Period and Cycle Length

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Desirable Generator Properties

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- ❑ The routine should be fast.
- ❑ Should not require a large amount of memory.
- ❑ It should have a long cycle or Large Period.
- ❑ The random values should be repeatable.
- ❑ The successive values should be independent and uniformly distributed.

Types of Random number Generators

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- ***Mid-Square method***, proposed by Von Neumann in the 1940's. The technique starts with a seed, the seed is squared and the middle digits become the random number.

Mid-Square Method

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□ Example:

□ $X_0 = 5497$

□ $X_0^2 = (5497)^2 = 30,217,009 \Rightarrow X_1 = 2170$

□ $R_1 = X_1 \times 1/10000 = 2170/10000 = 0.2170$ (it satisfies the condition for uniform distribution $U(0,1)$; 0.217 is between 0 and 1.

□ $X_1^2 = (2170)^2 = 04,708,900 \Rightarrow X_2 = 7089$

□ $R_2 = X_2 \times 1/10000 = 7089/10000 = 0.7089$ (it satisfies the condition for uniform distribution $U(0,1)$; 0.7089 is between 0 and 1. And the same process goes on for values of $X_3, X_4, \dots, X_n$.

□ **Problems:** difficult to assure that the sequence will not degenerate/deteriorate over a long period of time,(try 5197 as a seed).

Mid-product Generators

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- Similar to the mid-square technique. We select two seeds each containing the same number of digits. Period of this technique is typically longer than the mid-square technique.

- **Example:**

$$\bullet U_n = X_{n-2} \cdot X_{n-1}$$

- seeds ... $X_0' = 2938$, $X_0 = 7229$
- $U_1 = X_0' X_0 = (2938)(7229) = 21, 238, 802 \Rightarrow X_1 = 2388$
- $R_1 = 0.2388$ (calculated in same way as mid-square method)
- $U_2 = X_0 X_1 = (7229)(2388) = 17, 262, 852 \Rightarrow X_2 = 2628$
- $R_2 = 0.2628$ (calculated in same way as mid-square method)

Constant Multiplier Technique

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- Variation of the mid-product technique using a constant (**K**) multiplier.

- **Example:**

- seed = $X_0 = 7233$, constant = $K = 3987$

- $V_1 = (K)(X_0)$

- $V_1 = (3987)(7233) = 28,837,971 \Rightarrow X_1 = 8379$

- $R_1 = 0.8379$

- $V_2 = (K)(X_1)$

- $V_2 = (3987)(8379) = 33,407,073 \Rightarrow X_2 = 4070$

- $R_2 = 0.4070$

- $V_n = K \cdot X_{n-1}$

Additive Congruential Method

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- Requires a sequence of n numbers $X_1, X_2, \dots, X_n$ and produces an extension of the sequence using the relationship:
 - ▣ $X_i = (X_{i-1} + X_{i-n}) \bmod m, i = n+1, \dots$
- **Example:**
 - ▣ original sequence = 57, 34, 89, 92, 16, $n = 5, m = 100$
 - ▣ Where $X = \{X_1, X_2, X_3, X_4, X_5\}$
 - ▣ $X_6 = (X_5 + X_1) \bmod 100 = 73 \bmod 100 = 73$
 - ▣ $R_1 = X_6 / 100 = 0.73$
 - ▣ $X_7 = (X_6 + X_2) \bmod 100 = 107 \bmod 100 = 7$
 - ▣ $R_2 = X_7 / 10 = 0.7$

Mixed Linear Congruential Generators (M-LCG)

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- Initially proposed in the 1950's, produces a sequence of integers between 0 and $m-1$ using the relationship:
$$X_{i+1} = (aX_i + c) \bmod m \text{ \& }$$
$$R_i = X_i / m$$
- Where X_i is the seed, a is the multiplier, c is the increment and m is the modulus (all non-negative)
- The selection of a , c , m and the seed drastically effect the properties of the generator.
- The X 's produced are integers between 0 and $m-1$.

Mixed Linear Congruential Generators (M-LCG)

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Example: (seed = 27, $a = 17$, $c = 43$, $m = 100$)

- ▣ $X_0 = 27$
- ▣ $X_1 = (aX_0 + c) \bmod m$
- ▣ $X_1 = (17 \cdot 27 + 43) \bmod 100 = 502 \bmod 100 = 2$ (*2 is between 0 and $m-1$ i.e., 2 is between 0 and 99*).
- ▣ $R_1 = X_1/m = 2/100 = 0.02$
- ▣ $X_2 = (aX_1 + c) \bmod m$
- ▣ $X_2 = (17 \cdot 2 + 43) \bmod 100 = 77 \bmod 100 = 77$ (*77 is between 0 and $m-1$ i.e., 77 is between 0 and 99*).
- ▣ $R_2 = X_2/m = 77/100 = 0.77$

How to make reasonable choice for M-LCG parameters?

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- A linear congruential generator has full period (cycle length is m) if and only if the following conditions hold:
 - The only positive integer that exactly divides both m and c is 1; $\gcd(m, c) = 1$.
 - If q is a prime number that divides m , then q divides $a-1$; i.e., $q \mid m \Rightarrow q \mid (a-1)$.
 - If 4 divides m , then 4 divides $a-1$; i.e., $4 \mid m \Rightarrow 4 \mid (a-1)$.
- If all the above (3) conditions are satisfied we could say LCG will have a full cycle length.

Proper Choice of Parameters

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Example

Let us choose $m = 2^{13}$, $a = 2^1$, $c = 5$

$$m = 2^{13} = 8192$$

Condition 1

$$\left. \begin{array}{l} \gcd(m, c) = 1 \\ \gcd(8192, 5) = 1 \end{array} \right\} \text{satisfied}$$

Condition 2

Let $q = 2$, which is a prime number that divides m (8192) i.e., $q | m = 2 | 8192 = 4096$ Then $q | a - 1$ is also true i.e., $2 | 2^1 - 1 = 2 | 1 = 0$.
(satisfied)

Condition 3

$$\left. \begin{array}{l} 4 | m = 4 | 8192 = 2048, \text{ then} \\ 4 | a - 1 = 4 | 2^1 - 1 = 4 | 1 = 0 \end{array} \right\} \text{satisfied.}$$

Combined Generators (Home Reading Assignment)

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- It is possible to combine two or more generators to produce a “better” generator
 - ▣ Adding random numbers from two or more generators:
$$w_n = (x_n + y_n) \bmod m$$
 - ▣ Exclusive-or random numbers from two or more generators.
 - ▣ Shuffling - uses one random number sequence as an index to decide which of several numbers generated by a second sequence should be returned.

Pitfalls to Avoid in Random-Number Generation

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- ❑ A complex set of operations will not necessarily lead to random results - a sequence of operations where the final results are difficult does not necessarily mean the generator will pass a test for uniformity and independence.
- ❑ A single statistical test is not sufficient to show a generator is good. The generator may in fact fail other criteria.
- ❑ Avoid generators that have problems with certain seeds

Pitfalls to Avoid in Random-Number Generation

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- Accurate implementation is important - the ***period*** and ***randomness of generators*** are guaranteed only if the generation formula is accurately implemented without any overflow or truncation.
- Bits of successive words generated by a random number generator are not randomly distributed - special techniques are required to generate random bits efficiently.
- It is better to use an established generator that has been tested than to invent a new one.

References: Textbooks

1. *Based in part on materials by Dr. Jameel Ahmed (Dean EED/Director Riphah International University Campus in Sector I-14, Hajj Complex, Islamabad Pakistan)*
2. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
3. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
4. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*

Appendix

What is a Uniform Distribution?

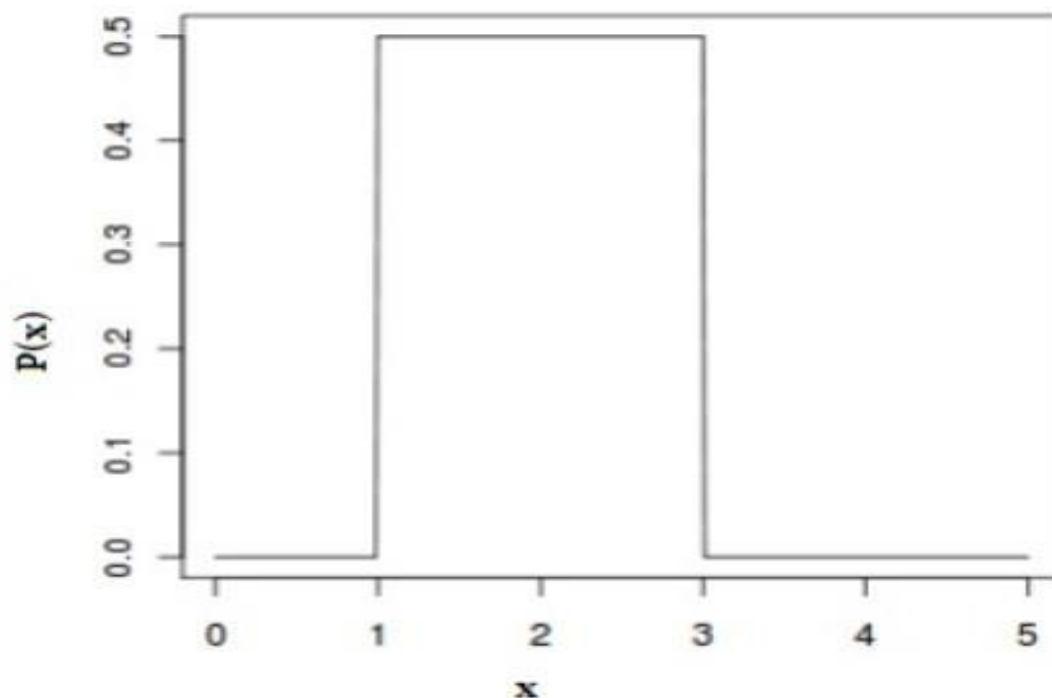
A **uniform distribution**, also called a **rectangular distribution**, is a probability distribution that has **constant probability**.

This distribution is defined by **two parameters**, a and b :

- a is the minimum.
- b is the maximum.

The distribution is written as $U(a, b)$.

The following graph shows the distribution with $a = 1$ and $b = 3$:



Relatively Prime Concept

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When two numbers have no common factors other than 1.

21 and 22 are relatively prime:

- The factors of 21 are 1, 3, 7 and 21
 - The factors of 22 are 1, 2, 11 and 22
- (the only common factor is 1)

But 21 and 24 are NOT relatively prime:

- The factors of 21 are 1, 3, 7 and 21
 - The factors of 24 are 1, 2, 3, 4, 6, 8, 12 and 24
- (the common factors are 1 AND 3)